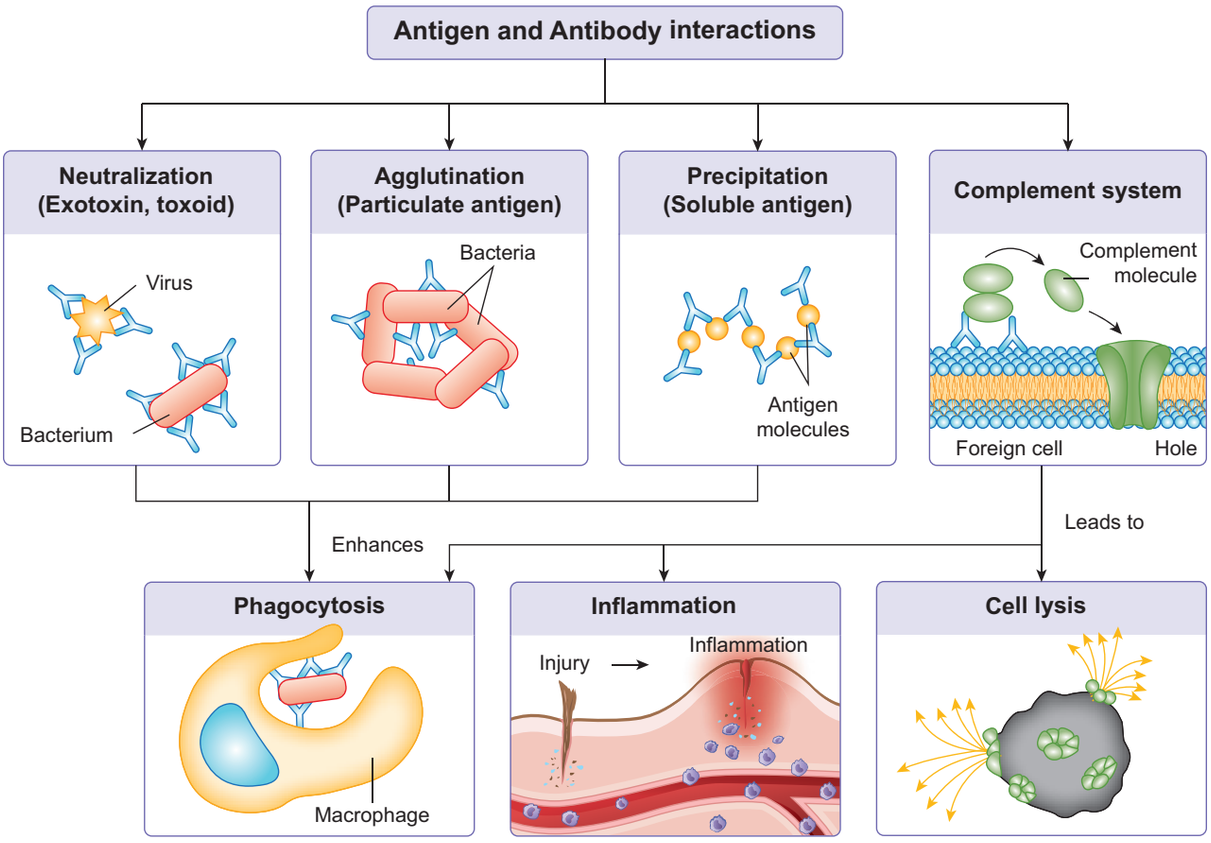


Fig. 7.16 Antigen and antibody reaction

Whenever a **particulate antigen** interacts with its antibody, it would result in **clumping or agglutination** of the particulate antigen, which is called **agglutination reaction**. The antibody involved in bringing about agglutination reaction is called **agglutinin**.

Opsonisation or enhanced attachment is the process by which a pathogen is marked of ingestion and destruction by a phagocyte. Opsonisation involves the binding of an **opsonin** i.e., antibody, to a receptor on the pathogen's cell membrane. After opsonin binds to the membrane, phagocytes are attracted to the pathogen. So, opsonisation is a process in which **pathogens are coated with a substance called an opsonin**, marking the pathogen out for destruction by the immune system. This results in a much more **efficient phagocytosis**.

The **neutralization** reactions are the reactions of antigen-antibody that involve the **elimination of harmful effects** of bacterial exotoxins or a virus by specific antibodies. These neutralizing substances i.e., antibodies are known as **antitoxins**. This specific antibody is produced by a host cell in response to a bacterial exotoxin or corresponding toxoid (inactivated toxin).

7.3.8 Vaccines

A vaccine **is a biological preparation that provides active acquired immunity** to a particular disease and resembles a disease-causing microorganism and is often made from weakened or attenuated or killed forms of the microbes, their toxins, or one of its surface proteins. Vaccines **"teach"** our body how to defend itself when viruses or bacteria, invade it. Vaccines deliver only very little amounts of inactivated or weakened viruses or bacteria, or parts of them. This allows the immune

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